

Proposition 4.40. *Let X be a connected CW complex with $H_1(X) = 0$. Then there is a simply-connected CW complex X^+ and a map $X \rightarrow X^+$ inducing isomorphisms on all homology groups.*

Proof: Choose loops $\varphi_\alpha: S^1 \rightarrow X^1$ generating $\pi_1(X)$ and use these to attach cells e_α^2 to X to form a simply-connected CW complex X' . The homology exact sequence

$$0 \rightarrow H_2(X) \rightarrow H_2(X') \rightarrow H_2(X', X) \rightarrow 0 = H_1(X)$$

splits since $H_2(X', X)$ is free with basis the cells e_α^2 . Thus we have an isomorphism $H_2(X') \approx H_2(X) \oplus H_2(X', X)$. Since X' is simply-connected, the Hurewicz theorem gives an isomorphism $H_2(X') \approx \pi_2(X')$, and so we may represent a basis for the free summand $H_2(X', X)$ by maps $\psi_\alpha: S^2 \rightarrow X'$. We may assume these are cellular maps, and then use them to attach cells e_α^3 to X' forming a simply-connected CW complex X^+ , with the inclusion $X \hookrightarrow X^+$ an isomorphism on all homology groups. $\square$

In the preceding proposition, the condition $H_1(X) = 0$ means that $\pi_1(X)$ is equal to its commutator subgroup, that is, $\pi_1(X)$ is a **perfect** group. Suppose more generally that X is a connected CW complex and $H \subset \pi_1(X)$ is a perfect subgroup. Let $p: \tilde{X} \rightarrow X$ be the covering space corresponding to H , so $\pi_1(\tilde{X}) \approx H$ is perfect and $H_1(\tilde{X}) = 0$. From the previous proposition we get an inclusion $\tilde{X} \hookrightarrow \tilde{X}^+$. Let X^+ be obtained from the disjoint union of $\tilde{X}^+$ and the mapping cylinder M_p by identifying the copies of $\tilde{X}$ in these two spaces. Thus we have the commutative diagram of inclusion maps shown at the right. From the van Kampen theorem, the induced map $\pi_1(X) \rightarrow \pi_1(X^+)$ is surjective with kernel the normal subgroup generated by H . Further, since X^+/M_p is homeomorphic to $\tilde{X}^+/\tilde{X}$ we have $H_*(X^+, M_p) = H_*(\tilde{X}^+, \tilde{X}) = 0$, so the map $X \rightarrow X^+$ induces an isomorphism on homology.

$$\begin{array}{ccc} \tilde{X} & \longrightarrow & \tilde{X}^+ \\ \downarrow & & \downarrow \\ X \simeq M_p & \longrightarrow & X^+ \end{array}$$

This construction $X \rightarrow X^+$, killing a perfect subgroup of $\pi_1(X)$ while preserving homology, is known as the Quillen **plus construction**. In some of the main applications X is a $K(G, 1)$ where G has perfect commutator subgroup, so the map $X \rightarrow X^+$ abelianizes π_1 while preserving homology. The space X^+ need no longer be a $K(\pi, 1)$, and in fact its homotopy groups can be quite interesting. The most striking example is $G = \Sigma_\infty$, the infinite symmetric group consisting of permutations of $1, 2, \dots$ fixing all but finitely many n 's, with commutator subgroup the infinite alternating group A_∞ , which is perfect. In this case a famous theorem of Barratt-Kahn-Priddy and Quillen says that the homotopy groups $\pi_i(K(\Sigma_\infty, 1)^+)$ are the stable homotopy groups of spheres!

There are limits, however, on which subgroups of $\pi_1(X)$ can be killed without affecting the homology of X . For example, for $X = S^1 \vee S^1$ it is impossible to kill the commutator subgroup of $\pi_1(X)$ while preserving homology. In fact, by Exercise 23

at the end of this section every space with fundamental group $\mathbb{Z} \times \mathbb{Z}$ must have H_2 nontrivial.

Fiber Bundles

A ‘short exact sequence of spaces’ $A \hookrightarrow X \rightarrow X/A$ gives rise to a long exact sequence of homology groups, but not to a long exact sequence of homotopy groups due to the failure of excision. However, there is a different sort of ‘short exact sequence of spaces’ that does give a long exact sequence of homotopy groups. This sort of short exact sequence $F \rightarrow E \xrightarrow{p} B$, called a *fiber bundle*, is distinguished from the type $A \hookrightarrow X \rightarrow X/A$ in that it has more homogeneity: All the subspaces $p^{-1}(b) \subset E$, which are called **fibers**, are homeomorphic. For example, E could be the product $F \times B$ with $p: E \rightarrow B$ the projection. General fiber bundles can be thought of as twisted products. Familiar examples are the Möbius band, which is a twisted annulus with line segments as fibers, and the Klein bottle, which is a twisted torus with circles as fibers.

The topological homogeneity of all the fibers of a fiber bundle is rather like the algebraic homogeneity in a short exact sequence of groups $0 \rightarrow K \rightarrow G \xrightarrow{p} H \rightarrow 0$ where the ‘fibers’ $p^{-1}(h)$ are the cosets of K in G . In a few fiber bundles $F \rightarrow E \rightarrow B$ the space E is actually a group, F is a subgroup (though seldom a normal subgroup), and B is the space of left or right cosets. One of the nicest such examples is the Hopf bundle $S^1 \rightarrow S^3 \rightarrow S^2$ where S^3 is the group of quaternions of unit norm and S^1 is the subgroup of unit complex numbers. For this bundle, the long exact sequence of homotopy groups takes the form

$$\cdots \rightarrow \pi_i(S^1) \rightarrow \pi_i(S^3) \rightarrow \pi_i(S^2) \rightarrow \pi_{i-1}(S^1) \rightarrow \pi_{i-1}(S^3) \rightarrow \cdots$$

In particular, the exact sequence gives an isomorphism $\pi_2(S^2) \approx \pi_1(S^1)$ since the two adjacent terms $\pi_2(S^3)$ and $\pi_1(S^3)$ are zero by cellular approximation. Thus we have a direct homotopy-theoretic proof that $\pi_2(S^2) \approx \mathbb{Z}$. Also, since $\pi_i(S^1) = 0$ for $i > 1$ by Proposition 4.1, the exact sequence implies that there are isomorphisms $\pi_i(S^3) \approx \pi_i(S^2)$ for all $i \geq 3$, so in particular $\pi_3(S^2) \approx \pi_3(S^3)$, and by Corollary 4.25 the latter group is $\mathbb{Z}$.

After these preliminary remarks, let us begin by defining the property that leads to a long exact sequence of homotopy groups. A map $p: E \rightarrow B$ is said to have the **homotopy lifting property** with respect to a space X if, given a homotopy $g_t: X \rightarrow B$ and a map $\tilde{g}_0: X \rightarrow E$ lifting g_0 , so $p\tilde{g}_0 = g_0$, then there exists a homotopy $\tilde{g}_t: X \rightarrow E$ lifting g_t . From a formal point of view, this can be regarded as a special case of the *lift extension property for a pair* (Z, A) , which asserts that every map $Z \rightarrow B$ has a lift $Z \rightarrow E$ extending a given lift defined on the subspace $A \subset Z$. The case $(Z, A) = (X \times I, X \times \{0\})$ is the homotopy lifting property.

A **fibration** is a map $p: E \rightarrow B$ having the homotopy lifting property with respect to all spaces X . For example, a projection $B \times F \rightarrow B$ is a fibration since we can choose lifts of the form $\tilde{g}_t(x) = (g_t(x), h(x))$ where $\tilde{g}_0(x) = (g_0(x), h(x))$.